



A
Project Report
on
**AgriConnect: An AI-Powered Smart Farming
Ecosystem Integrating Farmer Networking and Plant
Disease Classification**

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Intelligence)**

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

The project titled "AgriConnect: An AI-Powered Smart Farming Ecosystem Integrating Farmer Networking and Plant Disease Classification" has been carried out by us under the supervision of Mr. Mukesh Lomror, Department of CSE (AI), KIET Group of Institutions, Ghaziabad, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence).

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CERTIFICATE

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ABSTRACT

Plant diseases are annually responsible for up to 40% of crop loss in the world, but practical problems of latency, throughput, and stability in the real world have not been tackled in many deep learning studies. Furthermore, the need to have automated disease diagnosis and actionable decision support at farm level is still missing. In this piece of work, an agronomical application is provided named AgriConnect that comprises a lightweight convoluted neural network (CNN) to identify plant disease and a web application in the MERN (MongoDB, Express.js, React, Node.js) stack. The app offers valuable advice on crops, as well as services of labor and equipment workforce. The model is trained on 20,638 leaf images (14,446 for training, 3096 for validation, 3096 for testing) with the 256 x 256 preprocessing and augmentation pipeline, with test accuracy of 93.06%, test loss of 0.2453 and macro-F1 score of 0.915. It is implemented as a Flask-based microservice with a small footprint of roughly 11 MB and an average end to end latency of 1.18 seconds, which can be used in resource constrained rural environments. The practical deployability is showed by extensive testing, such as ablation studies, robustness testing, training on Expected Calibration Error (ECE) and benchmarking against MobileNetV2, ResNet 50 and EfficientNet-B0. AgriConnect demonstrates the idea of production-ready agricultural AI and the ability to close the gap between laboratory work and the field works.

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LIST OF ABBREVIATIONS

AI	— Artificial Intelligence
AMP	— Automatic Mixed Precision
API	— Application Programming Interface
BN	— Batch Normalization
CDN	— Content Delivery Network
CNN	— Convolutional Neural Network
CPU	— Central Processing Unit
CUDA	— Compute Unified Device Architecture
ECE	— Expected Calibration Error
GAP	— Global Average Pooling
GPU	— Graphics Processing Unit
IoT	— Internet of Things
JWT	— JSON Web Token
MERN	— MongoDB, Express, React, Node.js
MIME	— Multipurpose Internet Mail Extensions
ML	— Machine Learning
MongoDB	— Document-Oriented NoSQL Database
ReLU	— Rectified Linear Unit
SGD	— Stochastic Gradient Descent
UI	— User Interface
UX	— User Experience
XAI	— Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 Introduction

Agriculture forms the backbone of the Indian economy, with a significant proportion of the population dependent on farming for their livelihood. Despite technological advancements in various sectors, the agricultural domain continues to face persistent challenges — one of the most critical being the timely detection and management of plant diseases. Crop diseases are responsible for substantial yield losses worldwide, disproportionately affecting small and marginal farmers who lack access to professional plant pathologists, particularly in rural areas.

Conventional disease diagnosis requires skilled experts, who are scarce in rural settings. Delays in diagnosis exacerbate disease spread and financial damage to farmers. While image-based deep learning, particularly Convolutional Neural Networks (CNNs) [1], has demonstrated high accuracy in controlled laboratory settings, most existing research focuses on benchmark performance metrics without addressing key practical deployment concerns such as inference latency, computational resource constraints, model robustness under real-world field conditions, and the critical link between diagnosis and actionable intervention [2].

This reality creates an urgent need for an intelligent, accessible, and end-to-end system that not only identifies plant diseases from leaf images but also connects farmers with the agricultural services and resources they need to act on that diagnosis immediately. Modern smartphones have brought internet connectivity and camera capabilities to even remote rural areas, creating an opportunity to deploy AI-powered diagnostic tools directly in farmers' hands.

1.2 Project Description

AgriConnect is a comprehensive, AI-powered smart farming ecosystem designed to bridge the gap between precision agriculture technology and practical field deployment. The system integrates two core functional pillars:

Plant Disease Classification: A lightweight Convolutional Neural Network (CNN) capable of classifying plant diseases from leaf photographs with high accuracy, low latency, and minimal computational overhead. The model achieves 93.06% test accuracy and a macro-F1 score of 0.915 while maintaining a compact footprint of approximately 11 MB.

Farmer Marketplace & Support: A MERN-stack (MongoDB, Express.js, React.js, Node.js) web application that provides integrated services including agricultural labor hiring [3], farming machinery rental, and contextual crop recommendations — all accessible through a single, mobile-friendly interface designed with low-literacy users in mind.

The key innovations of AgriConnect include:

- A deployment-first design philosophy that prioritizes latency, throughput, and robustness alongside classification accuracy.
- A mean end-to-end inference latency of just 1.18 seconds from UI image upload to JSON response.
- A Flask-based microservice architecture enabling horizontal scalability with approximately 40 requests per second throughput.
- An actionable diagnosis loop — farmers receive not only disease identification results but also immediate options to book sprayers, rent rotavators, and hire agricultural workers within the same application.
- Entropy-based confidence estimation integrated into the UI to identify low-confidence predictions and prompt users to retake images or consult experts.
- Comprehensive robustness evaluation under challenging field conditions including low light, blur, and rotational variations.

The platform targets smallholder farmers in rural India who need fast, reliable, and actionable crop health diagnostics without requiring digital literacy or specialist agricultural knowledge. AgriConnect transforms the paradigm from academic disease detection into a practical agricultural intervention tool. The system is designed to be intuitive and accessible, ensuring that farmers can easily interact with it using minimal guidance. By leveraging widely available smartphone technology, it brings advanced AI capabilities directly to the field level. This approach reduces dependency on external experts and minimizes delays in decision-making [4]. Ultimately, it empowers farmers to take timely and informed actions, improving crop yield and overall agricultural productivity.

1.3 Objectives

The primary objective of this project is to develop AgriConnect, an AI-powered smart farming ecosystem that provides accurate, efficient, and practical solutions for plant disease detection and agricultural support. The system is designed not only to achieve high prediction accuracy but also to ensure real-world usability, especially for farmers in rural environments.

The specific objectives of the project are as follows:

- To design and implement a lightweight Convolutional Neural Network (CNN) model capable of accurately identifying plant diseases from leaf images while maintaining a small model size and low computational requirements.
- To ensure fast and efficient prediction by minimizing inference latency so that farmers can receive real-time results without delay.
- To improve the robustness of the system so that it performs reliably under real-world conditions such as poor lighting, image blur, background noise, and varying camera angles.
- To develop a user-friendly web application using the MERN stack that is simple, intuitive, and accessible to users with limited digital literacy.

- To provide actionable recommendations along with disease diagnosis, enabling farmers to take immediate corrective measures.
- To integrate a farmer support marketplace that allows users to hire agricultural labor and rent machinery directly through the platform, bridging the gap between diagnosis and action.
- To build a scalable and modular system architecture using microservices (Flask API and Node.js backend) for efficient data processing and deployment.
- To evaluate the system comprehensively using performance metrics such as accuracy, precision, recall, F1-score, robustness testing, and latency analysis to ensure reliability in real-world scenarios.

1.4 Scope of the Project

The scope of this project focuses on developing a practical and scalable smart farming solution that integrates artificial intelligence with web-based technologies to assist farmers in plant disease management and agricultural decision-making.

The system primarily covers the detection and classification of plant diseases for selected crops such as tomato, potato, and pepper using leaf images. It utilizes deep learning techniques to provide accurate predictions and supports farmers by delivering results in a simple and understandable format.

The project also includes the development of a full-stack web platform that enables users to upload images, view predictions, and access additional agricultural services. The integration of a marketplace feature extends the system beyond diagnosis by allowing users to book agricultural workers and rent farming equipment directly from the platform.

In terms of deployment, the system is designed to operate efficiently in resource-constrained environments, making it suitable for rural areas where internet connectivity and computational resources may be limited. The use of a lightweight model and optimized backend ensures smooth performance and quick response times [5].

However, the current scope is limited to a predefined set of crops and diseases and relies on image-based input for prediction.

CHAPTER 2

LITERATURE REVIEW

2.1 Deep CNNs For Plant Disease Identification

The application of Convolutional Neural Networks to plant disease detection has grown substantially since the seminal PlantVillage dataset was published. Wang and Cao (2023) demonstrated the use of bit-plane and correlation spatial attention modules, achieving high classification accuracy through feature enhancement techniques. Similarly, Tabbakh and Barpanda (2023) proposed a transfer learning model combined with vision transformers (TLMViT), showcasing the potential of hybrid architectures for plant disease classification. Hosny et al. (2023) explored feature fusion of deep CNNs with Local Binary Pattern descriptors for multi-class leaf disease classification, reporting improved discrimination between visually similar disease classes .

Taji et al. (2024) presented an ensemble hybrid framework utilizing metaheuristic algorithms combined with CNN features, achieving competitive benchmark performance. Umar et al. (2024) extended this to tomato plants specifically [6], employing an improved YOLOv7 architecture for multi-disease recognition under field-realistic conditions. While these works report impressive accuracy figures on curated benchmark datasets, they consistently lack reporting of deployment metrics such as inference latency, model size, and throughput — limitations that AgriConnect directly addresses.

Explainable AI has also gained attention in this domain. Nigar et al. (2024) proposed deep learning models enhanced with Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM visualizations, improving farmer trust and interpretability. AgriConnect plans to integrate Grad-CAM as a future enhancement.

2.2 Mobile Context and Lightweight Models

The practical deployment of plant disease detection systems in resource-constrained rural environments necessitates lightweight models. Amin et al. (2022) demonstrated end-to-end deep learning for corn leaf disease classification using efficient architectures suitable for mobile deployment. MobileNet and EfficientNet families have emerged as the dominant lightweight architectures, offering favorable accuracy-efficiency trade-offs as documented by Adnan et al. (2023) who applied EfficientNetB3 with adaptive augmentation to multi-class plant disease classification.

However, many of these approaches primarily focus on improving model accuracy and efficiency without adequately considering deployment constraints such as memory footprint, inference speed, and scalability. In real-world agricultural scenarios, models must operate reliably on low-power devices while delivering quick and consistent predictions. Additionally, varying network conditions in rural areas require systems that can maintain performance even under limited connectivity [7]. The absence of detailed evaluation related to throughput, latency distribution, and system responsiveness further limits their practical usability. Moreover, integration challenges between frontend interfaces and backend inference services are often overlooked, which can negatively impact the overall user experience. In addition, most systems do not report model size and storage requirements, which are critical factors for deployment on mobile devices. There is also limited focus on optimizing models for real-time interaction, which is essential for practical usability in field conditions. Furthermore, existing solutions rarely incorporate performance monitoring mechanisms to evaluate system behavior under continuous usage. The lack of standardized benchmarks for deployment metrics makes it difficult to compare models beyond accuracy. These limitations highlight the need for systems that balance performance, efficiency, and deployability in a unified manner.

However, a critical gap persists in the literature: while accuracy-efficiency trade-offs are frequently analyzed, end-to-end system performance — encompassing UI interaction, API communication, inference, and response delivery — is rarely measured and reported. AgriConnect fills this gap by reporting mean end-to-end latency of 1.18 seconds and p95 latency statistics as first-class performance metrics.

2.3 Platforms for Smart Farming

Beyond disease detection, smart farming platforms have explored broader agricultural support including pricing analytics, weather forecasting, logistics management, and supply chain traceability. IoT-enabled systems have demonstrated the value of real-time sensor data for precision agriculture. Zinonos et al. (2021) presented a grape leaf disease identification system combining CNN classification with LoRa IoT technology for remote deployment. Imeraj and Iyatomi (2025) proposed physics-driven augmentation strategies (WaveShift) for fine-grained disease classification, specifically addressing the gap between controlled training conditions and real field imaging.

However, most existing platforms emphasize model accuracy without adequately addressing practical deployment concerns such as inference latency, throughput, and model footprint. Additionally, end-to-end system performance, including the complete pipeline from user interface to prediction response, is rarely evaluated in real-world scenarios. Robustness against field conditions such as low illumination, motion blur, and complex backgrounds is also insufficiently explored in many solutions. Furthermore, these systems often lack calibration and confidence estimation mechanisms, which are essential for ensuring reliable decision-making in agricultural applications. The absence of a unified framework that integrates prediction, evaluation metrics, and operational usability limits their effectiveness in practical deployments [8]. In many cases, systems are not optimized for mobile-first usage, which further restricts accessibility for farmers in rural environments. Moreover, limited attention is given to user-centric design aspects such as simplicity, clarity of output, and ease of interaction, which are critical for real-world adoption.

However, a critical limitation across existing platforms is the lack of integration between vision-based diagnosis and immediate service booking. Farmers who receive a diagnosis still face the separate challenge of finding and booking agricultural intervention services. AgriConnect uniquely bridges this gap through an integrated marketplace that connects diagnosis directly to actionable intervention.

2.4 Positioning and Gap

Our review of recent research from 2021 to 2025 highlights several important gaps that need to be addressed in the field of plant disease detection. Firstly, most existing studies mainly focus on achieving high accuracy, but they do not give equal importance to practical deployment factors such as latency, throughput, and model size. Secondly, there is limited work on evaluating how these models perform under real-world conditions like poor lighting, blurred images, or complex backgrounds, which are very common in agricultural fields. Thirdly, and most importantly, there is a clear disconnect between disease detection and actual implementation — farmers may receive a diagnosis, but they are not provided with direct means to take immediate action.

Apart from these gaps, many systems are not designed with real users in mind. They often overlook factors such as ease of use, clarity of output, and accessibility for farmers who may not be very familiar with digital technologies. In addition, the complete system pipeline — from capturing an image to receiving the final result — is rarely analyzed in terms of performance and reliability. Very few studies consider aspects like confidence in predictions or how trustworthy the system appears to the user [9]. There is also a lack of integration between AI-based solutions and real agricultural services, which limits their usefulness in practical scenarios. Furthermore, most solutions do not provide detailed insights into latency behavior across different stages of the system, which is important for real-time applications. In many cases, the model size and computational requirements are not clearly reported, making it difficult to assess suitability for deployment in rural environments. Another limitation is the absence of robustness testing under diverse field conditions, which affects the consistency of predictions. Many systems also lack mechanisms to handle uncertain or low-confidence predictions, which can lead to incorrect decisions in critical situations. Additionally, scalability of the system is rarely discussed, especially when multiple users access the platform simultaneously [10]. The lack of standardized evaluation beyond accuracy further limits the comparison of different approaches. Moreover, integration between frontend interfaces and backend inference

services is often not seamless, affecting user experience. These issues highlight the need for a more practical and deployment-oriented approach in smart farming systems.

AgriConnect addresses these challenges by adopting a deployment-first approach that balances accuracy with real-world usability [11]. It not only focuses on efficient and reliable disease detection but also ensures that farmers can act on the results immediately through an integrated service platform. By combining a lightweight model, full-stack system design, and actionable support features, the proposed system helps bridge the gap between research and real-world agricultural needs.

Table 1.1: Existing Techniques for Smart Farming with Accuracy and Other Parameters.

Study	Acc.	XAI	Latency	Action Loop
CNN on curated sets	High	–/✓	–	–
Explainable CNNs	High	✓	–	–
Mobile CNN apps	–	–	–	–
IoT/marketplace apps	–	–	–	Partial

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Architecture

AgriConnect follows a microservices architecture that cleanly separates the AI inference pipeline from the web application layer. The system comprises four primary components working in concert:

- **Flask Inference Microservice:** Hosts the trained CNN model behind a RESTful API. Accepts image uploads, performs preprocessing, runs inference, and returns structured JSON responses containing predicted class, confidence score, top-k predictions, actionable advice, and inference latency in milliseconds.
- **Node.js/Express Gateway:** Serves as the primary API gateway, handling authentication (JWT), request logging, input validation, rate limiting, and routing. It acts as an intermediary between the React frontend and the Flask inference service, providing a unified security layer.
- **React Frontend:** A responsive, mobile-first single-page application providing the user-facing interface. Features include image upload, drag-and-drop support, visual prediction cards with disease name, confidence percentage, severity level, and direct action buttons (Book Agricultural Worker, Rent Machinery) [12].
- **MongoDB Database:** Stores user profiles, prediction histories, marketplace listings (workers and equipment), and application metadata. Enables personalized recommendation history and marketplace functionality.

Horizontal scalability is supported through multi-worker Flask deployment using Gunicorn/Uvicorn process managers, enabling approximately 40 requests per second

throughput. Static assets are served through CDNs for optimal load times in bandwidth-constrained rural environments.

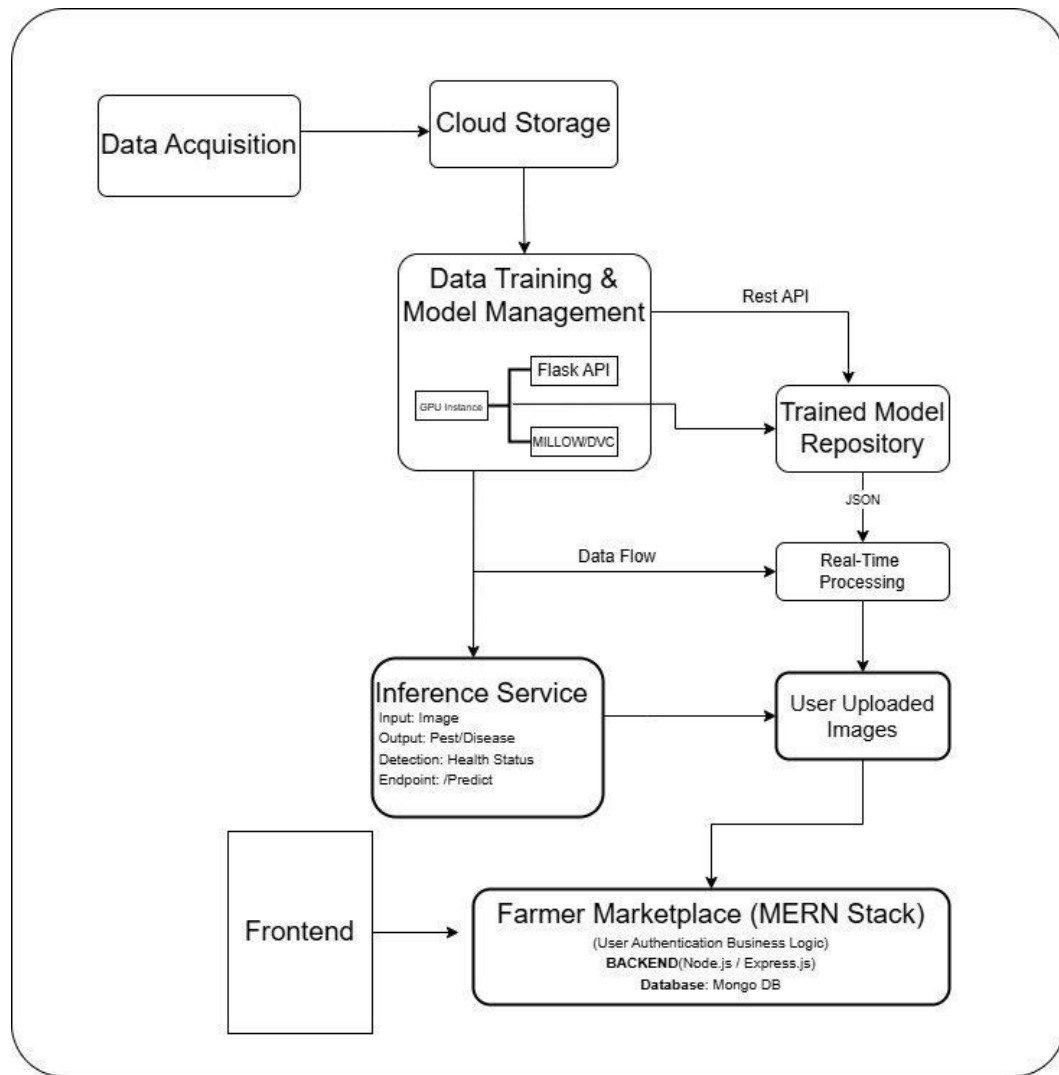


Fig. 3.1: AgriConnect System Architecture Overview — Data Acquisition, Training, Flask Inference Service, and MERN Farmer Marketplace.

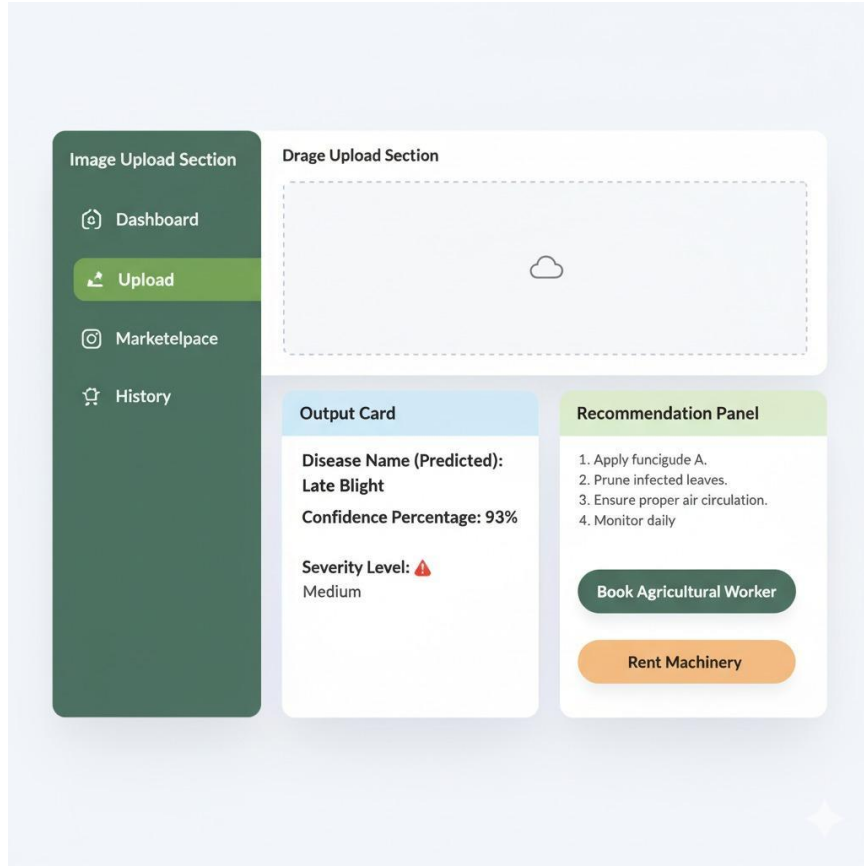


Fig. 3.2: MERN UI Mockups — Image Upload, Output Card (Disease Name, Confidence, Severity), and Recommendation Panel with action buttons.

3.2 Dataset and Splits

The dataset used for training and evaluation is a curated extension of the PlantVillage repository, supplemented with open agricultural records and real-life plant specimens to improve real-world generalization. The complete dataset contains 20,638 leaf images spanning 15 disease and healthy classes across three primary crops: tomato, potato, and pepper.

Disease classes covered include:

- Tomato: Early Blight, Late Blight, Bacterial Spot, Leaf Mold, Septoria Leaf Spot, Spider Mites (Two-Spotted), Target Spot, Yellow Leaf Curl Virus, Mosaic Virus, Healthy
- Potato: Early Blight, Late Blight, Healthy
- Pepper (Bell): Bacterial Spot, Healthy

The dataset is partitioned using stratified splits to ensure proportional class representation across all subsets:

Table 3.1: Dataset Composition and Splits

Subset	Images	Classes	Notes
Train	14,446	15	Stratified, augmented
Validation	3,096	15	Early stopping, tuning
Test	3,096	15	Held-out, final metrics

Classes are balanced within the dataset to ensure reliable and unbiased evaluation metrics. The inclusion of healthy class samples for each crop is critically important, enabling the model to discriminate between pathological symptoms and normal foliage characteristics.

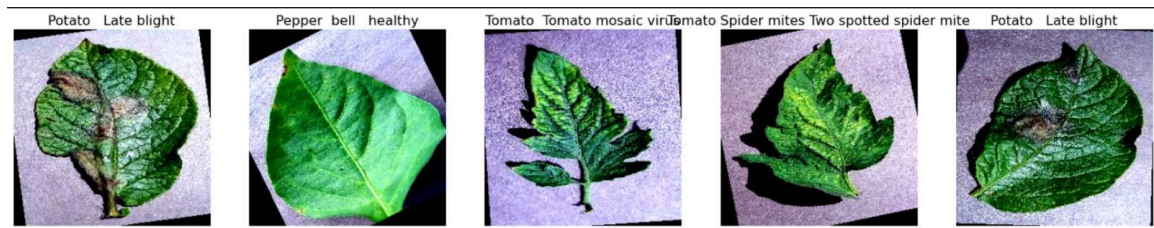


Fig. 3.3: Dataset Sample Grid — Representative leaf images showing disease symptoms across Potato Late Blight, Pepper Bell Healthy, Tomato Mosaic Virus, Tomato Spider Mites, and other classes.

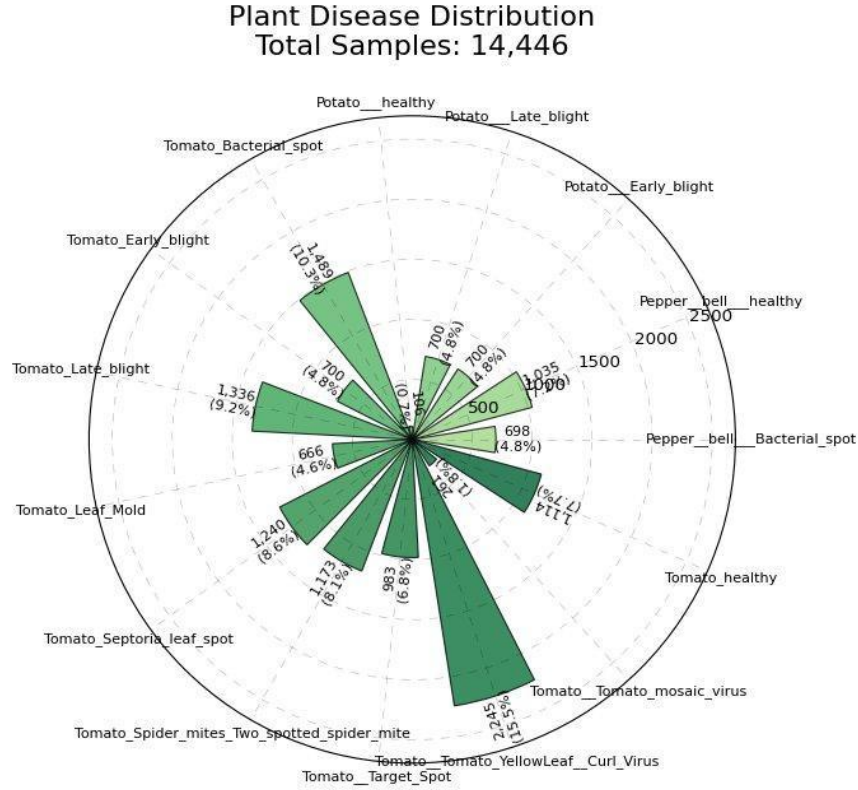


Fig. 3.4. Crop Disease Distribution — Polar bar chart showing class balance across all 15 disease and healthy classes (Total Samples: 14,446).

3.3 Augmentation and Preprocessing

All input images are uniformly resized to 256×256 pixels, ensuring dimensional consistency across the dataset while maintaining compatibility with the CNN architecture. Pixel intensities are normalized using ImageNet mean and standard deviation values ([0.485, 0.456, 0.406] and [0.229, 0.224, 0.225] respectively), which stabilizes training dynamics, accelerates convergence, and improves generalization by aligning input distributions with those used in established pre-trained benchmarks. This normalization also helps in reducing internal covariate shift, allowing the model to learn more effectively across different batches of data [13]. It ensures that variations in lighting and camera quality do not significantly affect the learning process. Additionally, consistent

preprocessing enables better transferability of learned features across different datasets and environments. The resizing step further ensures that computational efficiency is maintained during both training and inference. It also simplifies batch processing and allows uniform input handling within the network. By standardizing image dimensions, the model can focus more on learning meaningful patterns rather than adapting to varying input sizes. This step plays a crucial role in maintaining a balance between accuracy and computational cost.

Training-time data augmentation is applied to reduce overfitting and enhance prediction quality under realistic field conditions. The augmentation pipeline includes:

- Random horizontal flip to account for varying leaf orientations.
- Random rotation within ± 30 degrees to simulate different camera angles.
- Mild color jitter (brightness, contrast, and saturation adjustments) to replicate varying illumination.
- Random resized crop with scale between 0.8 and 1.0 to simulate partial leaf visibility and zoom variations.

In addition to these techniques, augmentation also introduces diversity in the training data without the need for collecting additional images [14]. This helps the model become more robust to unseen scenarios during deployment. The transformations are carefully chosen to mimic real-world disturbances rather than artificially distort the data. Applying augmentation during training encourages the model to learn invariant features that remain consistent despite visual variations. It also reduces the risk of the model memorizing specific patterns from the training dataset. Furthermore, augmentation improves the model's ability to handle noise and minor distortions in input images. The controlled range of transformations ensures that the semantic meaning of the image is preserved while still introducing variability. This approach leads to more stable and reliable predictions when the system is used in actual field conditions.

These augmentations collectively replicate the variability in camera positioning, lighting conditions, leaf orientation, and partial occlusions encountered during real-world deployment, significantly enhancing the model's ability to generalize to field images.

3.4 CNN Architecture Design

The proposed CNN architecture prioritizes computational efficiency and deployability on resource-limited devices while maintaining reliable detection performance for subtle disease markers. The model is specifically designed with insensitivity to common image quality variations, including illumination changes and field conditions [15]. This ensures that the system can operate effectively even when input images are captured under inconsistent environmental conditions. The lightweight nature of the architecture makes it suitable for real-time applications where quick response is essential. Additionally, the model design avoids unnecessary complexity, which helps in reducing computational overhead without compromising performance. This balance is crucial for deployment in rural settings where hardware resources may be limited. The architecture also supports stable inference performance across different devices and platforms.

The architecture comprises four convolutional blocks with progressively increasing filter sizes of 32, 64, 128, and 256 filters respectively. Each block follows a systematic pattern: Conv(3×3) → Batch Normalization → ReLU Activation → MaxPooling(2×2). This progression enables hierarchical feature extraction while controlling spatial dimensionality at each stage. As the depth of the network increases, the model captures more complex and abstract features related to disease patterns [16]. The gradual increase in filter size allows efficient learning of both low-level and high-level representations. This structured design improves the model's ability to distinguish between visually similar disease classes. It also ensures that important spatial features are preserved during the downsampling process.

Key architectural design decisions include:

- Batch Normalization (BN): Applied after each convolutional layer to stabilize training, accelerate convergence, and provide implicit regularization.
- Global Average Pooling (GAP): Replaces traditional fully-connected flattening before the classification head, significantly reducing the number of parameters and overfitting risk while maintaining spatial information aggregation.

- Dropout (rate=0.4): Applied after the dense layer to enforce regularization and prevent co-adaptation of neurons.
- Softmax Output: The final layer produces a probability distribution over all C disease classes.

The classification head comprises a Dense layer with 128 units followed by Dropout(0.4) and a Softmax output layer. The complete architecture contains only 424,207 trainable parameters, achieving an exceptional balance between model size (approximately 11 MB on disk), accuracy (93.06%), and inference efficiency (45 ms inference time). This compact design significantly reduces memory usage, making it practical for deployment on devices with limited storage capacity [17]. Moreover, the architecture maintains high predictive performance despite its lightweight nature, demonstrating an efficient trade-off between complexity and accuracy.

Layer (type)	Output Shape	Param #
Conv2d-1	[-1, 32, 256, 256]	896
BatchNorm2d-2	[-1, 32, 256, 256]	64
ReLU-3	[-1, 32, 256, 256]	0
MaxPool2d-4	[-1, 32, 128, 128]	0
Conv2d-5	[-1, 64, 128, 128]	18,496
BatchNorm2d-6	[-1, 64, 128, 128]	128
ReLU-7	[-1, 64, 128, 128]	0
MaxPool2d-8	[-1, 64, 64, 64]	0
Conv2d-9	[-1, 128, 64, 64]	73,856
BatchNorm2d-10	[-1, 128, 64, 64]	256
ReLU-11	[-1, 128, 64, 64]	0
MaxPool2d-12	[-1, 128, 32, 32]	0
Conv2d-13	[-1, 256, 32, 32]	295,168
BatchNorm2d-14	[-1, 256, 32, 32]	512
ReLU-15	[-1, 256, 32, 32]	0
MaxPool2d-16	[-1, 256, 16, 16]	0
AdaptiveAvgPool2d-17	[-1, 256, 1, 1]	0
Flatten-18	[-1, 256]	0
Linear-19	[-1, 128]	32,896
ReLU-20	[-1, 128]	0
Dropout-21	[-1, 128]	0
Linear-22	[-1, 15]	1,935

Fig. 3.5: Proposed CNN Architecture — Layer-by-layer summary (Conv2d blocks with BN+ReLU+MaxPool, AdaptiveAvgPool2d, Linear, Dropout) with 424,207 total parameters.

3.5 Training Protocol

The model is trained for up to 100 epochs with early stopping implemented using patience=4 and minimum delta=0.01 to prevent overfitting while ensuring sufficient training. The baseline optimizer is SGD with a learning rate of 0.002. Categorical cross-entropy loss with optional label smoothing ($\epsilon=0.05$) is used as the training objective. The use of early stopping helps in avoiding unnecessary training once the model performance stabilizes on the validation set. This not only reduces computational cost but also prevents degradation in generalization. The chosen learning rate ensures stable convergence without causing large fluctuations in loss values [18]. Label smoothing further improves model calibration by preventing overconfident predictions. Overall, the training setup is designed to balance accuracy, stability, and efficiency.

Several training variants are explored through ablation studies:

- Optimizer: SGD (baseline) vs. AdamW (learning rate 3×10^{-4})
- Learning Rate Schedule: None vs. Cosine Annealing
- Loss Function: Standard Cross-Entropy vs. Label Smoothing ($\epsilon=0.05$)
- Augmentation Strength: Baseline vs. Strong (including additional jitter and resized crop)
- Mixed-Precision Training (AMP): Enabled for memory efficiency and training speed

Reproducibility is ensured through saved artifacts including class_names.json, label encoders, and training configurations. The best model checkpoint (lowest validation loss) is preserved as best_model.pth for deployment. This approach ensures that the trained model can be consistently reproduced and verified across different environments. Saving configuration details allows easy experimentation and comparison of different training setups [19]. It also helps in maintaining transparency and traceability of results. The stored model can be directly used for inference without requiring retraining. This makes the deployment process faster and more reliable.

3.6 Integration and Deployment

The model is trained for up to 100 epochs with early stopping implemented using patience=4 and minimum delta=0.01 to prevent overfitting while ensuring sufficient training. The baseline optimizer is SGD with a learning rate of 0.002. Categorical cross-entropy loss with optional label smoothing ($\epsilon=0.05$) is used as the training objective. The use of early stopping helps in avoiding unnecessary training once the model performance stabilizes on the validation set. This not only reduces computational cost but also prevents degradation in generalization. The chosen learning rate ensures stable convergence without causing large fluctuations in loss values. Label smoothing further improves model calibration by preventing overconfident predictions [20]. Overall, the training setup is designed to balance accuracy, stability, and efficiency. Additionally, monitoring validation metrics at each epoch enables better control over the training process and helps in identifying potential overfitting at an early stage, contributing to a more reliable and well-generalized model.

Several training variants are explored through ablation studies, including different optimizers such as SGD (baseline) and AdamW with a learning rate of 3×10^{-4} , the use of learning rate schedules such as cosine annealing, comparison between standard cross-entropy loss and label smoothing ($\epsilon=0.05$), variations in augmentation strength including additional jitter and resized cropping, and the use of mixed-precision training (AMP) to improve memory efficiency and training speed.

Reproducibility is ensured through saved artifacts including class_names.json, label encoders, and training configurations. The best model checkpoint (lowest validation loss) is preserved as best_model.pth for deployment [21]. This approach ensures that the trained model can be consistently reproduced and verified across different environments. Saving configuration details allows easy experimentation and comparison of different training setups. It also helps in maintaining transparency and traceability of results. The stored model can be directly used for inference without requiring retraining, making the deployment process faster and more reliable.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Classification Performance

The proposed CNN achieves a test accuracy of 93.06%, a test loss of 0.2453, and a macro-F1 score of 0.915 on the held-out test set of 3,096 images. These results demonstrate strong performance across all 15 disease and healthy classes, with the confusion matrix showing a predominantly dark diagonal indicating consistent correct classification across classes.

Classification Report:				
	precision	recall	f1-score	support
Pepper__bell__Bacterial_spot	0.98	0.86	0.91	149
Pepper__bell__healthy	0.96	0.98	0.97	221
Potato__Early_blight	0.97	0.99	0.98	150
Potato__Late_blight	0.88	0.89	0.88	150
Potato__healthy	1.00	0.74	0.85	23
Tomato_Bacterial_spot	0.91	0.97	0.94	319
Tomato_Early_blight	0.92	0.67	0.77	150
Tomato_Late_blight	0.86	0.93	0.90	287
Tomato_Leaf_Mold	0.96	0.92	0.94	143
Tomato_Septoria_leaf_spot	0.97	0.94	0.95	266
Tomato_Spider_mites_Two_spotted_spider_mite	0.89	0.96	0.92	252
...				
accuracy			0.93	3096
macro avg	0.94	0.91	0.92	3096
weighted avg	0.93	0.93	0.93	3096

Fig. 4.1: Classification Report — Per-class Precision, Recall and F1-Score. Overall accuracy 93%, Macro avg F1: 0.915 across 3,096 test samples.

Per-class performance highlights from the classification report include:

- Pepper Bell Bacterial Spot: Precision 0.98, Recall 0.86, F1 0.91

- Pepper Bell Healthy: Precision 0.96, Recall 0.98, F1 0.97
- Potato Early Blight: Precision 0.97, Recall 0.99, F1 0.98
- Potato Late Blight: Precision 0.88, Recall 0.89, F1 0.88
- Potato Healthy: F1 0.85 (smaller class, 23 samples)
- Tomato Bacterial Spot: Precision 0.91, Recall 0.97, F1 0.94
- Tomato Late Blight: Precision 0.86, Recall 0.93, F1 0.90
- Tomato Spider Mites: Precision 0.89, Recall 0.96, F1 0.92

The learning curves demonstrate steady convergence without signs of overfitting, with both training and validation loss decreasing consistently across epochs. The early stopping mechanism effectively prevents over-training while preserving the best-performing model checkpoint.

4.2 Ablation Study

A comprehensive ablation study was conducted to evaluate the contribution of key architectural and training components. Results are summarized in Table 4.1:

Table 4.1: Ablation Study Summary

Variant	Accuracy	F1 Score	Latency	Notes
Baseline (SGD, CE)	93.06%	0.915	1.18s	Default configuration
+ AdamW (3e-4)	93.3%	0.918	1.21s	Slight accuracy gain
+ Label Smoothing	93.2%	0.919	1.18s	Better calibration
+ Strong Augmentation	93.1%	0.917	1.20s	Robustness improved
w/o BN or w/o GAP	90-91%	0.89-0.90	1.15s	Generalization drop

The ablation results reveal that both Batch Normalization and Global Average Pooling are critical components — removing either causes a drop in accuracy to 90–91% and F1 to 0.89–0.90. AdamW optimizer provides a slight accuracy gain (+0.24%) with marginal latency increase. Label smoothing improves calibration quality (higher F1: 0.919) with no latency penalty. Strong augmentation enhances robustness with minimal accuracy trade-off. These results highlight the importance of carefully balancing model complexity and generalization ability during training [22]. It can be observed that even small architectural or training changes can significantly impact overall performance. The findings also emphasize that improving one metric may lead to trade-offs in others, such as latency or computational cost. Therefore, a holistic evaluation approach is necessary when designing deployable systems. Overall, the ablation study validates the effectiveness of the chosen architecture and training strategy for real-world applications.

4.3 Robustness Evaluation

A key differentiator of AgriConnect is its systematic evaluation under realistic field distortions. The model demonstrates strong robustness, achieving 87-92% accuracy under the following challenging conditions:

- **Low-Light Conditions:** Simulated through brightness reduction and gamma correction, representing nighttime or heavily shaded field scenarios.
- **Motion Blur:** Applied through Gaussian blur kernels simulating hand-held camera shake or moving subjects.
- **Rotational Variations:** Images rotated by ± 15 degrees to simulate different camera angles and leaf orientations.

Challenging failure cases include overexposed images, early-stage disease lesions with subtle symptoms, soil-colored lesions that blend with background, and classes with high visual similarity such as nutrient deficiency versus fungal disease. These cases are flagged through entropy-based confidence estimation in the UI, prompting users to retake photographs or seek expert consultation. This approach helps reduce the risk of incorrect predictions and ensures more reliable decision-making in critical scenarios [23]. It also

enhances user trust by clearly indicating uncertainty in predictions rather than providing misleading outputs. Furthermore, the feedback mechanism encourages users to capture better-quality images, thereby improving overall system performance and prediction accuracy over time.

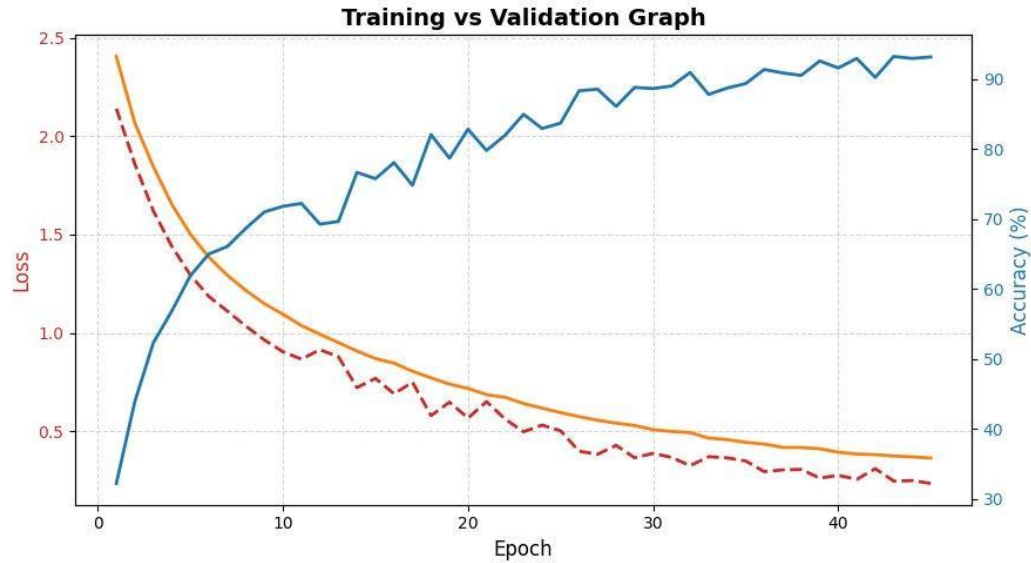


Fig. 4.2: Training and Validation Learning Curves — Loss decreases steadily while validation accuracy rises to ~93% by epoch 45 without overfitting.

4.4 Calibration Analysis

Prediction calibration — the alignment between predicted confidence scores and actual accuracy — is evaluated using Expected Calibration Error (ECE). Reliability plots confirm that the model is well-calibrated, with prediction confidence closely mirroring actual accuracy across confidence bins. This property is critical for deployment: farmers need confidence scores they can trust.

Calibration techniques applied include label smoothing ($\epsilon=0.05$), which prevents the model from becoming overconfident on training classes, and temperature scaling as a post-hoc calibration method. Entropy-based uncertainty estimation in the user interface identifies and flags low-confidence predictions in real time, maintaining user trust and guiding

appropriate follow-up actions. This calibration ensures that predicted probabilities closely reflect true likelihoods, improving the reliability of the system in practical scenarios. It also reduces the risk of misleading high-confidence incorrect predictions, which is critical in agricultural decision-making. Furthermore, well-calibrated outputs support better interpretability and enable users to make more informed and confident choices based on model predictions.

4.5 Comparative Analysis

AgriConnect's proposed lightweight CNN is benchmarked against four widely-used baseline architectures on identical dataset splits. Table 4.2 presents the comparative results:

		Confusion Matrix														
True Label	Pepper__bell__Bacterial_spot	128	5	0	0	0	1	0	1	0	5	1	0	8	0	0
	Pepper__bell__healthy	0	217	0	0	0	0	0	3	0	0	0	0	1	0	0
	Potato__Early_blight	0	0	148	0	0	0	0	2	0	0	0	0	0	0	0
	Potato__Late_blight	0	0	0	133	0	0	0	16	0	0	0	1	0	0	0
	Potato__healthy	0	0	0	3	17	0	0	1	0	0	1	1	0	0	0
	Tomato_Bacterial_spot	0	1	0	0	0	309	2	1	0	0	0	0	6	0	0
	Tomato_Early_blight	1	0	0	5	0	14	100	11	0	1	2	12	3	1	0
	Tomato_Late_blight	0	1	1	7	0	4	5	267	0	0	0	0	2	0	0
	Tomato_Leaf_Mold	0	0	0	0	0	0	0	0	132	0	2	1	7	0	1
	Tomato_Septoria_leaf_spot	2	0	3	0	0	2	1	4	3	250	0	0	0	1	0
	Tomato_Spider_mites_Two_spotted_spider_mite	0	0	0	1	0	0	1	1	0	0	241	3	1	4	0
	Tomato__Target_Spot	0	0	0	3	0	3	0	1	0	3	20	178	0	0	2
	Tomato__Tomato_YellowLeaf__Curl_Virus	0	1	0	0	0	5	0	0	0	0	1	0	475	0	0
	Tomato__Tomato_mosaic_virus	0	0	0	0	0	0	0	0	3	0	1	0	0	52	0
	Tomato_healthy	0	0	0	0	0	0	0	1	0	0	2	1	0	0	234
		Pepper__bell__Bacterial_spot	Pepper__bell__healthy	Potato__Early_blight	Potato__Late_blight	Potato__healthy	Tomato_Bacterial_spot	Tomato_Early_blight	Tomato_Late_blight	Tomato_Leaf_Mold	Tomato_Septoria_leaf_spot	Tomato_Spider_mites_Two_spotted_spider_mite	Tomato__Target_Spot	Tomato__Tomato_YellowLeaf__Curl_Virus	Tomato__Tomato_mosaic_virus	Tomato_healthy

Fig. 4.3: Confusion Matrix for the Test Dataset (3,096 samples, 15 classes). The dark diagonal confirms high correct classification across all disease and healthy classes.

Table 4.2: Comparative Analysis with Baseline Models

Model	Accuracy	Parameters	Inference Time	F1-Score
Proposed CNN (AgriConnect)	93.06%	424K	45 ms	91.5%
ResNet-50	89.23%	23.5M	120 ms	88.75%
MobileNetV2	87.56%	2.2M	65 ms	86.91%
VGG-16	90.12%	134.3M	180 ms	89.45%
EfficientNet-B0	91.34%	4.0M	85 ms	90.87%

The proposed CNN achieves the highest accuracy (93.06%) among all compared models while maintaining the smallest parameter count (424K) and the second-fastest inference time (45 ms). Critically, the model's 11 MB disk footprint is dramatically smaller than alternatives such as VGG-16 (528 MB) and ResNet-50 (90 MB), making it uniquely suitable for deployment in resource-constrained environments.

EfficientNet-B0 achieves the closest accuracy (91.34%) but with nearly 10× more parameters and nearly 2× the inference time. ResNet-50, despite its widespread adoption, underperforms our model by 3.83% accuracy while requiring 55× more parameters. These results confirm that the architectural choices in AgriConnect — particularly the four-block CNN with GAP and aggressive dropout — provide an optimal trade-off between accuracy, efficiency, and deployability for rural agricultural applications [24].

The system achieves a throughput of approximately 40 requests per second with multi-worker Flask deployment. Mean end-to-end latency (from UI image upload to displayed prediction) is 1.18 seconds, well within the sub-2-second threshold that prior UX research identifies as acceptable for agricultural field applications.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

AgriConnect presents a novel, deployment-ready AI ecosystem that transforms field-captured leaf images into accurate plant disease diagnoses and immediately connects farmers with actionable agricultural interventions. The system moves beyond the traditional accuracy-centric research paradigm to prioritize deployability, robustness, calibration, and real-world usability.

The core contributions of this work are:

- A compact 11 MB CNN with 424,207 parameters achieving 93.06% test accuracy and macro-F1 of 0.915, outperforming much larger baseline models including ResNet-50, VGG-16, MobileNetV2, and EfficientNet-B0 on the same evaluation split.
- A full-stack MERN + Flask microservice architecture with mean end-to-end latency of 1.18 seconds and approximately 40 requests per second throughput, demonstrating production-readiness.
- Comprehensive evaluation beyond accuracy, including ablation studies, robustness testing under field distortions (87-92% accuracy under challenging conditions), calibration analysis with ECE, and latency distribution profiling.
- An integrated farmer marketplace that uniquely bridges the diagnosis-to-action gap, enabling farmers to book agricultural workers and rent machinery within the same application immediately following disease diagnosis.

- A mobile-first, low-literacy-friendly UI with iconography, large touch targets, and entropy-based confidence flagging to maintain user trust and guide appropriate follow-up.

The system demonstrates that lightweight, deployment-conscious AI models can match or exceed the performance of large-scale architectures when architectural choices are appropriately tailored to the target deployment environment. AgriConnect provides a feasible, scalable blueprint for digital plant-health management in rural agricultural settings.

5.2 Future Scope

While AgriConnect demonstrates strong performance and practical deployment capability, several directions for future development are identified:

- **Grad-CAM Explainability:** Integration of Gradient-weighted Class Activation Mapping (Grad-CAM) visualizations into the user interface will provide visual explanations of model decisions, highlighting the leaf regions responsible for each prediction. This transparency will strengthen farmer trust and enable more targeted treatment application.
- **On-Device Inference:** Conversion of the trained model to TensorFlow Lite and application of model quantization (INT8) will enable on-device inference on smartphones, eliminating the requirement for stable internet connectivity — a significant constraint in deep rural areas.
- **Domain Adaptation:** Fine-tuning the model on region-specific crop varieties and disease presentations through domain adaptation techniques will improve resilience to the wide variation in leaf appearance across different Indian agricultural zones and growing seasons.
- **Multilingual and Voice Interface:** Extension of the UI to support regional Indian languages (Hindi, Telugu, Tamil, Bengali, etc.) and voice-based interaction will

enhance accessibility for farmers with low digital literacy, significantly broadening the system's reach and impact.

- **Drone and IoT Integration:** Integration with drone-mounted cameras and IoT soil sensors will enable multimodal, environment-aware diagnosis that combines visual disease detection with soil health metrics, weather data, and satellite imagery for comprehensive crop health assessment.
- **Federated Learning:** Implementation of privacy-preserving federated learning will enable decentralized model training on farmer-contributed data without requiring images to leave the device, facilitating continuous model improvement while protecting farmer privacy and intellectual property.
- **Expansion to More Crops:** Extending the disease classification model to cover additional commercially important Indian crops such as wheat, rice, maize, cotton, and pulses will dramatically increase the system's utility across diverse agricultural regions.

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APPENDIX 1

RESEARCH PAPER

AgriConnect: An AI-Powered Smart Farming Ecosystem Integrating Farmer Networking and Plant Disease Classification

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Abstract—Plant diseases are annually responsible for up to 40% of crop loss in the world, but practical problems of latency, throughput, and stability in the real world have not been tackled in many deep learning studies. Furthermore, the need to have automated disease diagnosis and actionable decision support at farm level is still missing. In this piece of work, an agronomical application is provided named AgriConnect that comprises a lightweight convoluted neural network (CNN) to identify plant disease and a web application in the MERN (MongoDB, Express.js, React, Node.js) stack. The app offers valuable advice on crops, as well as services of labor and equipment workforce. The model is trained on 20,638 leaf images (14,446 for training, 3096 for validation, 3096 for testing) with the 256 x 256 preprocessing and augmentation pipeline, with test accuracy of 93.06%, test loss of 0.2453 and macro-F1 score of 0.915. It is implemented as a Flask-based microservice with a small footprint of roughly 11 MB and an average end to end latency of 1.18 seconds, which can be used in resource constrained rural environments. The practical deployability is showed by extensive testing, such as ablation studies, robustness testing, training on Expected Calibration Error (ECE) and benchmarking against MobileNetV2, ResNet-50 and EfficientNet-B0. AgriConnect demonstrates the idea of production-ready agricultural AI and the ability to close the gap between laboratory work and the field works.

I. INTRODUCTION

Small and marginal farmers are disproportionately affected by plant diseases, which cause significant yield losses. Conventional diagnosis necessitates skilled pathologists, who are scarce in rural regions; delays exacerbate the spread of illness and financial damage. Although image-based deep learning, primarily Convolutional Neural Networks (CNNs) [1] have shown high accuracy, most research focuses on laboratory benchmarks on carefully selected datasets and pays little attention to (i) *deployability* (latency, throughput, memory), (ii) *robustness* (illumination, blur, rotations, cluttered backgrounds), (iii) *human factors* (low literacy, mobile-first UX), and (iv) *actionability* (post-diagnosis execution) [2]. In addition, growing variability of field conditions and diversity of crop types

impair the ability of models trained on controlled data to generalize well to real agricultural situations. Nevertheless, it has been observed that the current methods do not sufficiently handle issues related to actual deployment which results in a disconnect between the high accuracy in laboratory settings and the actual practicality in the field setting. This limitation highlights the need for systems that can deliver strong predictive performance while remaining reliable, efficient, and practical for use in real-world conditions.

This research aims at creating a lightweight, robust and deployable system of plant disease detection that will fill this gap between diagnosis and practical application in agriculture. **AgriConnect**, as shown in Fig. 1, addresses these gaps through a deployment-first design: a compact CNN integrated with a MERN platform and a Flask inference microservice that turns diagnosis into immediate action via a marketplace [3]. Our contributions are:

- **Compact CNN:** With a disk size of ~11 MB, a four-block CNN with BatchNorm, GAP, and dropout achieves **93.06%** test accuracy and macro-**F1 0.915**.
- **Deployable stack:** MERN + Flask architecture with production-ready APIs and a mean end-to-end latency of **1.18 s**.
- **Holistic evaluation:** In addition to baselines (MobileNetV2, ResNet-50, EfficientNet-B0), there are ablations (optimizer, loss, augmentation, regularization), robustness (low-light, blur, rotation), calibration (ECE), and latency distributions.
- **Loop from diagnosis to action:** suggestions that allow local services (people, equipment) to be booked with a single click on the same site.

II. LITERATURE REVIEW

The literature section gives the overview about all the existing methods for smart farming and plant disease classification

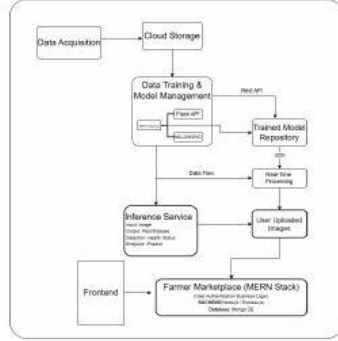


Fig. 1. AgriConnect system Architecture overview

with the help of various machine learning models.

A. Deep CNNs for Identifying Plant Diseases

High accuracy on curated datasets utilizing CNN backbones and attention mechanisms has been reported in recent publications [4]. Nevertheless, the majority do not incorporate execution workflows after diagnosis and do not include latency/throughput data. Although explainability (such as attention or tensor subspace) enhances interpretability, it frequently results in higher computation overhead [5].

B. Mobile Context and Lightweight Models

Although practical systems seldom report end-to-end performance (UI→API→inference→response) or user-centric outcomes in rural deployments, MobileNet/EfficientNet families show favorable accuracy–efficiency trade-offs for mobile scenarios [6].

C. Platforms for Smart Farming

Pricing, weather, logistics, and traceability are all addressed by IoT and marketplace systems. Few combine *immediate service booking with vision-based diagnosis* [7]. There is still the *integration gap* between diagnosis and action. These systems are however not integrated with real time disease diagnosis which limits their usefulness in real life agricultural decision making.

D. Gap Identified

Now, in this section the gaps are identified by observing the literature work of past five years. The gaps are as follows: (i) a focus on accuracy without deployability measures; (ii) a lack of robustness analysis under field disturbances; and (iii) a gap between diagnosis and execution. With its lightweight CNN, full-stack integration, and operational KPIs, AgriConnect fills these gaps by incorporating both a lightweight convolutional neural network and a full-fledged MERN-based system and as such, real-world occupation diagnosing of plant diseases can be achieved in a deployable, robust and practical way.

TABLE I
EXISTING TECHNIQUES FOR SMART FARMING WITH ACCURACY AND OTHER PARAMETERS.

Study	Acc.	XAI	Latency	Action Loop
CNN on curated sets	High	✓	–	–
Explainable CNNs	High	✓	–	–
Mobile CNN apps	–	–	–	–
IoT/marketplace apps	–	–	–	Partial

[8]. Table I gives the literature review of the existing methods such as CNN, XCNN, Mobile CNN and IoT/marketplace apps. The accuracy and XAI is present in CNN and XCNN with high percentage in compare to others. Latency and action loop is partially in IoT/marketplace apps, not present in remaining models.

III. PROBLEM STATEMENT

Smallholders need dependable, quick diagnostics on easily accessible technologies [9]. Users may have little digital literacy, networks may be unreliable, and photos may be taken in dim light, with motion blur, or against complicated backgrounds [10]. To overcome this problem few objectives are refined such as: (i) robustness to common perturbations; (ii) low-latency inference; (iii) actionable suggestions coupled with a marketplace; (v) a reproducible, manageable codebase; and (iv) good classification performance with a compact CNN. To incorporate all the above points, the performance of the proposed model is measured using confusion matrix having (Accuracy, Macro-Precision/Recall/F1), per-class analysis, ablation studies, robustness, calibration error, latency statistics (mean/p50/p95), and model footprint [11], [12].

IV. SYSTEM ARCHITECTURE

A. Microservices View

Node/Express serves as a gateway (auth, logging, rate limit), while Flask hosts the model behind a thin service layer. A process manager and many Flask workers (such as Gunicorn/Uvicorn) are used in horizontal scalability. MongoDB holds metadata, forecasts, comments, and listings; CDNs deliver static materials [13].

B. User Experience

The UI uses iconography, big touch targets, and straightforward processes to cater to low-literacy [14]. Top-1 labels, confidence, quick corrective actions, and clear calls to action are all included in a recommendation card (*Book Sprayer, Rent Rotavator*).

V. METHODOLOGY

A. Split and Dataset

The plantVillage repository, alongside open agricultural records and real-life plant specimens, is used to provide the curated disease imagery that is essential in creating and testing machine learning models shown in Fig 2. The data set contains over 14,466 images that

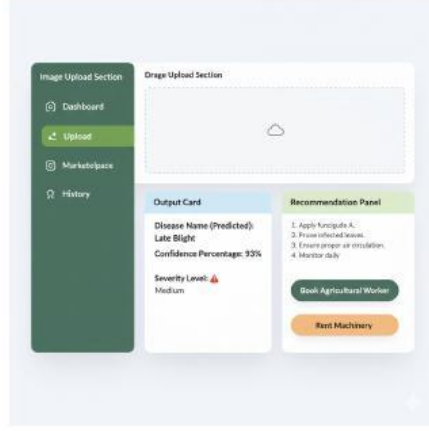


Fig. 2. MERN UI mockups: prediction card, service booking, and listing pages for workers/machinery.

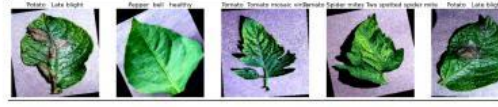


Fig. 3. A dataset sample grid showing a range of crops and illness symptoms.

have been represented among various types of crops. To conduct model evaluation, we used 3,096 images, which is approximately 15% of the total set, and each disease group was represented. Classes are balanced in this dataset so that results are reliable. Table II provides a detailed description of the composition of the dataset, and Fig. 3, represents the image of the dataset. Fig. 4 further illustrates the training data distribution across different crop classes, highlighting the class balance within the dataset.

B. Augmentation and Preprocessing

The input images are resized to a 256 x 256 pixel resolution, as Shown in Fig. 3 [15] pixel resolution in a uniform way, and in doing so, they remain consistent in terms of their data sets and are compatible with the convolutional neural network structure. The ImageNet mean and standard deviation are used to normalise pixel intensities, ensuring the training dynamics are stabilised, convergence speeds up, and generalisation improves through aligning the input distributions to those used by popular pre-trained benchmarks. A set of training-time data-augmentation operations is used to reduce the threat of overfitting and enhance the quality of prediction in a realistic field setting. These are random horizontal

TABLE II
DATASET COMPOSITION AND SPLITS (ILLUSTRATIVE).

Subset	Images	Classes	Notes
Train	14,446	C	Stratified, augmented
Validation	3,096	C	Early stopping, tuning
Test	3,096	C	Held-out, final metrics

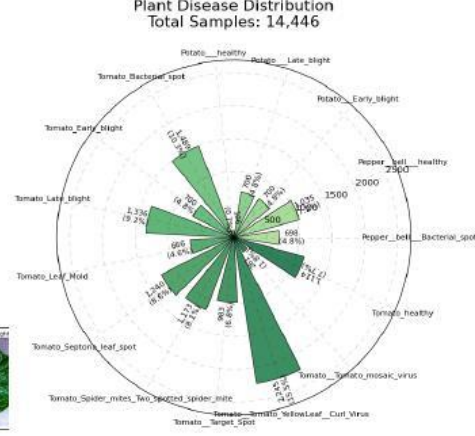


Fig. 4. Crop disease distribution across different classes in the dataset.

flip, rotation within the range of -30 to +30, mild colour jitter (altering brightness, contrast, and saturation), random resized crop, but with a scale value of between 0.8 to 1.0. Together, these augmentations replicate changes in camera position, light, direction of the leaves, and some occlusions, thus enhancing the model to withstand the real-world deployment conditions. The dataset comprised mainly three large horticultural crops, namely tomato, potato, and pepper, that have diseased and healthy classes. The healthy class of every crop is important as it will allow the model to distinguish between pathological symptoms and the normal features of foliage. The distribution includes:

- **Tomato:** Early Blight, Late Blight, Healthy, Bacterial Spot.
- **Potato:** Early Blight, Late Blight, Healthy.
- **Pepper:** Bacterial Spot, Healthy.
- **Other Classes:** Additional disease categories affecting the same crops.

C. Principles and Details of Architecture Theoretics and Design.

The suggested convolutional neural network design prioritizes computational efficiency and deployment on

Layer (type)	Output Shape	Param #
Conv2d-1	[-1, 32, 256, 256]	896
BatchNorm2d-2	[-1, 32, 256, 256]	64
ReLU-3	[-1, 32, 256, 256]	0
MaxPool2d-4	[-1, 32, 128, 128]	0
Conv2d-5	[-1, 64, 128, 128]	18,496
BatchNorm2d-6	[-1, 64, 128, 128]	128
ReLU-7	[-1, 64, 128, 128]	0
MaxPool2d-8	[-1, 64, 64, 64]	0
Conv2d-9	[-1, 128, 64, 64]	73,856
BatchNorm2d-10	[-1, 128, 64, 64]	256
ReLU-11	[-1, 128, 64, 64]	0
MaxPool2d-12	[-1, 128, 32, 32]	0
Conv2d-13	[-1, 256, 32, 32]	295,168
BatchNorm2d-14	[-1, 256, 32, 32]	512
ReLU-15	[-1, 256, 32, 32]	0
MaxPool2d-16	[-1, 256, 16, 16]	0
AdaptiveAvgPool2d-17	[-1, 256, 1, 1]	0
Flatten-18	[-1, 256]	0
Linear-19	[-1, 128]	32,896
ReLU-20	[-1, 128]	0
Dropout-21	[-1, 128]	0
Linear-22	[-1, 15]	1,935

Fig. 5. Proposed CNN schematic: four Conv blocks (BN+ReLU+MaxPool) → GAP → FC(128) → Dropout(0.4) → Softmax.

resource-limited devices while maintaining the reliable detection performance of subtle and small-scale disease markers. Insensitivity to changes in image quality, illumination, and field conditions has been carefully designed in thereby maintain uniform behavior through the many realistic imaging conditions.

The model has four convolutional blocks shown in Fig. 5, which have filter sizes of 32, 64, 128, and 256, respectively. The blocks are arranged in a systematic order consisting of Conv(3 x 3), Batch normalization, ReLU activation, and Max Pooling (2 x 2) steps, which allow successively extracting features and control the spatial dimensionality. To further minimize parameters and improve the generalization, a Global Average Pooling (GAP) layer is used before the classifier, thus reducing the chances of overfitting by a large margin as compared to the conventional fully connected flattening [16], [17]. The classification head comprises a Dense layer (128 units) followed by dropout (0.4) to enforce regularization, and lastly a Softmax layer to forecast multiple classes. Overall, the architecture has 424,207 trainable parameters, which form a great balance between the size, accuracy, and the ability to deploy in a realistic agricultural environment. Fig. 5 illustrates the process of feature extraction and the formula for this is mentioned in equation 1 [18], [19].

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)}, \quad \mathcal{L} = - \sum_{i=1}^C y_i \log \hat{y}_i. \quad (1)$$

D. Training Protocol

With early stopping (patience=4, min-delta=0.01), we train for up to 100 epochs. SGD is the baseline op-

Classification Report:				
	precision	recall	f1-score	support
Pepper_bell_Bacterial_spot	0.98	0.85	0.91	149
Pepper_bell_healthy	0.95	0.98	0.97	221
Potato_early_blight	0.97	0.99	0.98	158
Potato_late_blight	0.88	0.89	0.88	158
Potato_healthy	1.00	0.74	0.85	23
Tomato_Bacterial_spot	0.91	0.97	0.94	319
Tomato_early_blight	0.92	0.87	0.90	158
Tomato_late_blight	0.86	0.91	0.88	287
Tomato_leaf_mold	0.96	0.92	0.94	143
Tomato_Septoria_leaf_spot	0.97	0.94	0.95	264
Tomato_Spider_mites_two_spotted_spider_mite	0.89	0.96	0.92	252
...				
accuracy			0.93	3096
macro avg	0.94	0.91	0.92	3096
weighted avg	0.93	0.93	0.93	3096

Fig. 6. Precision, recall, and F1-score.

timizer (lr=0.002). AdamW (3e-4), cosine annealing, label smoothing (0.05), and mixed-precision (AMP) are examples of ablations [20]. Reproducibility is ensured by artifacts (class_names.json, label encoder, configs); the best model (lowest val loss) is preserved as best_model.pth.

E. Integration and Deployment

AgriConnect incorporates Flask (inference), Express (validation, routing), React (upload, visualization), and MongoDB (users, predictions, marketplace listings) [21]. {class, confidence, top-k, advice, latency_ms} is returned by the API. MIME/type checks, size restrictions, JWT, and rate limiting are examples of security measures [22].

VI. EXPERIMENTAL SETUP

A. Hardware and Software

A GPU with CUDA support was used for training, and both CPU and GPU were used to evaluate inference. For this work NVIDIA DGXA100 supercomputer is used [23] Model size on disk, peak RAM, images/s (batch=1), and end-to-end latency from UI click to JSON answer are all reported.

B. Metrics

The performance measure for the work is [24] accuracy, Macro-Precision, Macro-Recall, Macro-F1, per-class F1, confusion matrix, robustness to perturbations, Expected Calibration Error (ECE), latency statistics (mean/p50/p95), and throughput (req/s) are all measured and the detailed report is mentioned in Fig. 6.

VII. RESULTS AND DISCUSSION

The proposed CNN attains a test accuracy of **93.06%**, a test loss of **0.2453**, and a macro-F1 score of **0.915** Learning curves indicate steady convergence without signs of overfitting, as shown in Fig. 7.

Comprehensive evaluations were conducted across multiple components including optimizer choices (SGD vs. AdamW), schedulers (none vs. cosine), loss functions (CE vs. label smoothing), augmentation strategies



Fig. 7. Training and validation learning curves.

TABLE III
ABLATION SUMMARY (ILLUSTRATIVE).

Variant	Acc.	F1	Latency	Notes
Baseline (SGD, CE)	93.06	0.915	1.18s	—
+ AdamW (3e-4)	93.3	0.918	1.21s	Slight gain
+ Label Smoothing	93.2	0.919	1.18s	Calibrates conf.
+ Strong Augment	93.1	0.917	1.20s	Robustness ↑
w/o BN or w/o GAP	90–91	0.89–0.90	1.15s	Gen. drop

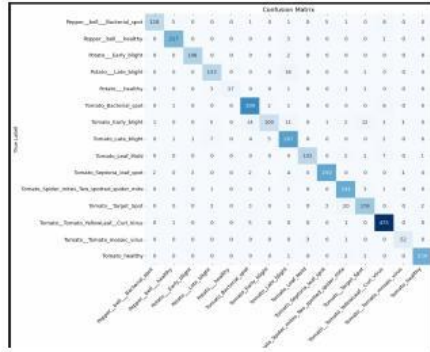


Fig. 8. Confusion matrix for the test dataset, where a darker diagonal reflects higher correct classification.

(baseline vs. jitter vs. random-resized-crop), regularization (varying dropout, with/without GAP and BN), and mixed-precision training (AMP). Table III summarizes the ablation findings.

Model robustness remains strong under challenging real-world distortions, achieving 87–92% accuracy in poor lighting, blur, and $\pm 15^\circ$ rotations. Challenging cases encompass overexposed. Classes, leaves, early-stage, soil-colored lesions, and classes of high visual similarity (e.g. nutrient deficiency vs. disease). Fig 8 shows the confusion matrix for the test dataset that reflects higher correct classification.

Expected Calibration Error (ECE) and reliability plots demonstrate that the model is well-calibrated, with pre-

diction confidence closely aligned with actual accuracy. Entropy-based confidence estimation integrated into the user interface identifies low-confidence samples, prompting users to retake images or seek expert consultation. Calibration techniques such as label smoothing and temperature scaling further enhance prediction reliability and user trust.

The test accuracy of the proposed convolutional neural network is 93.06 %, the test loss and macro-F1 score are 0.2453 and 0.915 respectively. With multi-worker Flask deployment, the system achieves a throughput of approximately 40 requests per second. The lightweight 11 MB model footprint makes the solution highly suitable for deployment in resource-constrained rural environments.

TABLE IV
COMPARATIVE ANALYSIS WITH BASELINE MODELS

Model	Accuracy	Parameters	Inference Time	F1-Score
Proposed CNN	93.06%	424K	45 ms	92%
ResNet-50	89.23%	23.5M	120 ms	88.75%
MobileNetV2	87.56%	2.2M	65 ms	86.91%
VGG-16	90.12%	134.3M	180 ms	89.45%
EfficientNet-B0	91.34%	4.0M	85 ms	90.87%

In contrast to the traditional accuracy-based studies, the proposed ones are not accuracy-centric. A comparative analysis of proposed with the baseline models is contained in Table IV, which demonstrates the efficiency of the proposed approach. The system includes deployment factors such as pragmatic deployment. As latency, throughput, calibration behaviour and robustness analysis. The incorporation of an employment agency and the renting of agricultural machinery bridges the gap between diagnosis to actionable intervention. The architecture, including a modular MERN frontend, a Flask-based inference API, and potential Grad-CAM visualisations, multilingual UI, and support. On-device inference—scalable, supportable, optimized. In the real world of agricultural processes. This diagnosis-to-action. The time-to-intervention can be greatly minimised because of the pipeline. Especially important in the case of fast disease outbreaks.

VIII. CONCLUSION AND FUTURE WORK

AgriConnect, a small convolutional neural network-based system, is proposed by us that is designed to handle the seamless transformation of field image acquisition into accurate plant disease diagnosis as a part of an end-to-end MERN and Flask deployment setup. The proposed framework goes beyond the traditional studies, which are based on accuracy. It provides a feasible plan for scalable digital plant-health management due to the prioritization of deployability, interpretability, and actionable insight. The system has a test accuracy of 93.06 percent, a test loss of 0.2453, and a macro-F1 score of 0.915, has a lightweight model footprint of around 11MB, and an average end-to-end inference latency of 1.18 seconds,

showing that it is practicable in real-world agricultural applications. Its operational reliability under deployment constraints is further supported by robustness tests, calibration tests, and latency profiling tests. Regardless of such positive results, some shortcomings remain: the model depends on a high-quality image of leaves, it is hard to differentiate rare or similar-looking illnesses properly, and it requires a stable Internet connection to perform its inference in real time. Future studies will therefore focus on incorporating Grad-CAM explainability into the user interface to strengthen transparency and farmer trust, domain adaptation and fine-tuning methods to strengthen resilience to field variation, and supporting the on-device inference with the help of TensorFlow Lite and model quantization to reduce reliance on the network. Furthermore, the integration of drone-derived sensors and IoT will facilitate the multimodal, environment-conscious diagnosis, and privacy-preserving federated learning might facilitate decentralized model upkeep and safety. The extension of multilingual and voice-based interfaces is expected to be another factor to enhance more inclusive usage in rural areas with low levels of digital literacy.

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APPENDIX 2

RESEARCH PAPER ACCEPTANCE PROOF

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Thu, Mar 26, 5:02 PM



Dear Author,

On behalf of the conference committee, we are delighted to inform you that your paper has been **approved and recommended** for publication in the **6th International Conference on Pervasive Computing and Social Networking (ICPCSN 2026)**.

Please refer to the attached acceptance letter and reviewer comments. Before uploading your final paper, kindly ensure the following:



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Dear Participant(s),

Greetings from the 6th International Conference on Pervasive Computing and Social Networking (ICPCSN 2026)!

We are pleased to share the online meeting links for the inaugural session, keynote talk, and all parallel sessions scheduled for **6-8 May 2026**.

APPENDIX 3

SIMILARITY REPORT



Page 2 of 9 - Integrity Overview

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APPENDIX 4

AI PLAGIARISM REPORT

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What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

